

LAGRANGE 插值实验报告



代码:

```
#include <stdio.h>
#include <math.h>
#define m 8
double lagrange
(double x[],double y[],double x0,int n)
{
    double Ln=0,l;
    for (int i=0;i<n;i++){
        l=1;
        for (int j=0;j<m;j++){
            if(j!=i){
                l=l*(x0-x[j])/(x[i]-x[j]);
            }
        }
        Ln=Ln+l*y[i];
    }
    return Ln;
}

int main()
{
    double x[]={0.5,0.7,0.9,1.1,1.3,1.5,1.7,1.9};
    double y[]={0.48,0.64,0.78,0.89,0.96,1.00,0.99,0.95};
    double a[]={0.74,1.6,0.55,1.2,1.85};
    double Ln[]={0,0,0,0,0};
    int n=8;
    for (int i=0;i<5;i++)
    {
        Ln[i]=lagrange(x,y,a[i],n);
        printf("Ln[%6.3f]= %12.8e\n", a[i], Ln[i]);
    }
    return 0;
}
```

运行结果：

```
D:\新建文件夹 (5)\大三\数值分析\lagrange.exe
Ln[ 0.740]= 6.68735821e-001
Ln[ 1.600]= 1.00341309e+000
Ln[ 0.550]= 5.26938820e-001
Ln[ 1.200]= 9.29135742e-001
Ln[ 1.850]= 9.52078056e-001

-----
Process exited after 0.2248 seconds with return value 0
请按任意键继续. . .
```